
Started on	Sunday, 12 February 2023, 8:25 PM
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Completed on	Sunday, 12 February 2023, 8:25 PM
Time taken	10 secs
Marks	0/81
Grade	0 out of 100

Question 1

Not answered

Marked out of 1

What is the prevalence of schizophrenia in people with intellectual disability?

Select one:

- ☐ 30%
- ☐ 15%
- ☐ 1%
- ☐ 3%
- ☐ 25%

Your answer is incorrect.

Explanation:

Schizophrenia is approximately 3 times more common in those with intellectual disability than in the general population. Most studies report a prevalence of around 3% for schizophrenia.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 826.

The correct answer is: 3%

Question 2

Not answered

Marked out of 1

Angelman syndrome is characterised by

Select one:

- ☐ Average intelligence
- ☐ Severe/profound intellectual disability
- ☐ Moderate intellectual disability
- ☐ Above average intelligence
- ☐ Mild intellectual disability

Your answer is incorrect.

Explanation:

Angelman syndrome is characterised by severe/profound intellectual disability. Approximately 86% of individuals have epilepsy.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 808.

The correct answer is: Severe/profound intellectual disability



Question 3

Not answered

Marked out of 1

Which of the following statements about Fragile X syndrome is not correct?

Select one:

- ☐ It accounts for 10-12% of intellectual disability in males
- ☐ It is associated with trinucleotide CGG repeats at a fragile site on the Y chromosome
- ☐ Macroorchidism is a clinical feature
- ☐ It is associated with mitral valve prolapse
- ☐ Brain imaging may show reduced posterior cerebellar vermis

Your answer is incorrect.

Explanation:

Fragile X syndrome is a common inherited cause of intellectual disability (ID) with X-linked dominant transmission. Penetrance is low, but greater in males than females due to the 'protective' effects of the second normal X chromosome in females. Fragile X syndrome accounts for 10-12% of ID in men. The syndrome is associated with a large sequence of triplet repeats (CGG) at a fragile site on the X chromosome (Xq27.3). The gene sequence is designated FMR-1. Clinical features include a large head, large ears, elongated face, high arched palate, short stature, mitral valve prolapse, poor muscle tone, large testes (macroorchidism), and hyperextensible joints. Brain imaging may show enlarged ventricles, enlarged hippocampus and caudate nuclei, and reduced posterior cerebellar vermis.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 814.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

The correct answer is: It is associated with trinucleotide CGG repeats at a fragile site on the Y chromosome

Question 4

Not answered

Marked out of 1

Autism is most commonly associated with

Select one:

- ☐ Intellectual disability
- ☐ Conduct disorder
- ☐ Schizophrenia
- ☐ Tourette syndrome
- ☐ ADHD

Your answer is incorrect.

Explanation:

80% of patients with autism have mild to moderate intellectual disability. The cause of autism is unknown, but a number of hypotheses exist: genetic (in Down syndrome and Fragile X); obstetric complications; toxic agents; pre/post-natal infections (with maternal rubella); autoimmune; and association with neurological disorders (e.g. tuberous sclerosis).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 822.

The correct answer is: Intellectual disability

Question 5

Not answered

Marked out of 1

Asperger syndrome is characterised by delays in

Select one:

- ☐ Nonverbal communication
- ☐ Receptive language
- ☐ Spoken language
- ☐ Cognitive development
- ☐ All of the listed options

Your answer is incorrect.

Explanation:

People with Asperger syndrome show no language delay and their cognitive development is not marked by an overall delay but by specific impairments in certain areas such as the executive functions. Children with autism spectrum disorder of an Asperger syndrome type often have difficulty deciphering the subtleties of non-verbal communication such as tone of voice, gestures, facial expressions, and body language.

Ref: Mirkovic B, Gérardin P. Asperger's syndrome: What to consider?. L'encephale. 2019 Apr 1;45(2):169-74.

The correct answer is: Nonverbal communication

Question 6

Not answered

Marked out of 1

Which of the following features is seen in hands of people with Down syndrome?

Select one:

- ☐ Altered dermatoglyphics
- ☐ Incurving of fingers
- ☐ Short, broad hands with a single palmar crease
- ☐ Webbed fingers
- ☐ All of the listed options

Your answer is incorrect.

Explanation:

Clinical features in the hands of people with Down syndrome include short, broad hands with a single palmar crease, syndactyly (webbed fingers), clinodactyly (incurving of fingers), and altered dermatoglyphics.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 806.

The correct answer is: All of the listed options

Question 7

Not answered

Marked out of 1

A 5-year-old boy with severe intellectual disability has a round face with hypertelorism, slanting palpebral fissures, broad flat nose, and an infantile cat-like cry. Which of the following conditions is he most likely to suffer from?

Select one:

- ☐ Cri-du-chat syndrome
- ☐ Down syndrome
- ☐ Angelman syndrome
- ☐ Smith-Magenis syndrome
- ☐ Williams syndrome

Your answer is incorrect.

Explanation:

Cri-du-chat syndrome is a partial monosomy; karyotype 5p- (varies from deletion of a small band at 5p15.2 to the entire arm of 5p). The infant has a characteristic high-pitched cry that resembles a cat. Clinical features include microcephaly, rounded face, hypertelorism, micrognathia, dental malocclusion, epicanthic folds, low-set ears, hypotonia, and severe/profound intellectual disability.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 808.

The correct answer is: Cri-du-chat syndrome

Question 8

Not answered

Marked out of 1

What is the prevalence of depression among subjects with a learning disability?

Select one:

- ☐ 10-15%
- ☐ 0.2%
- ☐ 1%
- ☐ 2-4%
- ☐ 0.35%

Your answer is incorrect.

The prevalence of depression is around 2-4%. Depressed mood and vegetative symptoms are the most striking symptoms, even though complaints of depression are not always expressed. A depressive mood is not verbalized, particularly among individuals at a lower intelligence level, but may well be observable.

The correct answer is: 2-4%

Question 9

Not answered

Marked out of 1

Which of the following is classified in ICD-11 as a developmental learning disorder?

Select one:

- ☐ Intellectual disability
- ☐ ADHD
- ☐ Autistic spectrum disorder
- ☐ Gerstmann syndrome
- ☐ Dyslexia

Your answer is incorrect.

Explanation:

In ICD-11, developmental learning disorder with impairment in reading (dyslexia) is characterised by significant and persistent difficulties in learning academic skills related to reading, such as word reading accuracy, reading fluency, and reading comprehension. In ICD-11, other developmental learning disorders can be diagnosed with impairment in written expression, with impairment in mathematics, and with other specified impairment of learning.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

Samuel D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 678.

The correct answer is: Dyslexia

Question 10

Not answered

Marked out of 1

Prior to the introduction of DSM-5, a multi-axial system was used by DSM-IV. DSM-IV coded mental retardation on which of the following axes?

Select one:

- ☐ Axis 4
- ☐ Axis 5
- ☐ Axis 1
- ☐ Axis 2
- ☐ Axis 3

Your answer is incorrect.

Explanation:

DSM-IV was a multi-axial diagnostic system, using five axes. Axis 2 coded any personality disorder and any mental handicap. This multi-axial approach was abandoned in DSM-5.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 11.

The correct answer is: Axis 2

Question 11

Not answered

Marked out of 1

The criteria for clinically significant challenging behaviour would include all except

Select one:

- ☐ Behaviour requires intervention by staff at least once a week
- ☐ Behaviour has caused more than minor injuries at some times
- ☐ Behaviour has caused over a few minutes disruption on a weekly basis
- ☐ Behaviour has caused over a few minutes' disruption at least daily
- ☐ Behaviours that caused damage that could not be rectified

Your answer is incorrect.

Criteria for clinically significant challenging behaviour is given by – 1. At some time, the behaviour has caused more than minor injuries to themselves or others, or destroyed their immediate living or working environment. 2. At least weekly the behaviours require intervention by staff; placed them in physical danger; caused damage that could not be rectified; caused at least 1hr of disruption. 3. Behaviour has caused over a few minutes' disruption at least daily. 4. Severe mental retardation and challenging behaviours: designing high-quality services (Chapman and Hall, London. Qureshi H (1994) The size of the problem)

The correct answer is: Behaviour has caused over a few minutes disruption on a weekly basis

Question 12

Not answered

Marked out of 1

Which of the following categories of learning disability is more commonly associated with disruptive behaviours and misconduct than the others listed ?

Select one:

- ☐ Severe learning disability
- ☐ Mild learning disability
- ☐ Borderline learning disability
- ☐ Profound learning disability
- ☐ Moderate learning disability

Your answer is incorrect.

Epidemiological surveys indicate that up to two-thirds of children and adults with mental retardation have comorbid mental disorders. The prevalence of psychopathology seems to be correlated with the severity of mental retardation; the more severe the mental retardation, the higher the risk for other mental disorders which would include mood disorders, schizophrenia, attention-deficit/hyperactivity disorder (ADHD), and conduct disorder. Disruptive and conduct disorder behaviours occurred more commonly in the mildly retarded group. Those with severe mental retardation have a particularly high rate of autistic disorder and pervasive developmental disorders.

The correct answer is: Mild learning disability

Question 13

Not answered

Marked out of 1

A 25-year-old gentleman with severe learning disability presents to the outpatient clinic with his live-in carer who mentioned that he indulges in repetitive self-injurious behaviour for no apparent reasons. There is no evidence of clinical depression. Which drug is particularly helpful in this situation?

Select one:

- ☐ Naltrexone
- ☐ Beta blockers
- ☐ Cryrosterone acetate
- ☐ Buspirone
- ☐ Methylphenidate

Your answer is incorrect.

Opiate antagonists (e.g. Naltrexone) may be effective in the treatment of repetitive self-injury. It may decrease self-injurious behaviour acutely, but it is less effective in the long term.

The correct answer is: Naltrexone



Question 14

Not answered

Marked out of 1

A 50-year-old man with Down syndrome is seen in the intellectual disability clinic. He may be developing Alzheimer disease. Which of the following tests would be most useful in establishing a diagnosis?

Select one:

- ☐ Dementia Questionnaire for People with Learning Disabilities
- ☐ Mini Mental State Examination
- ☐ Addenbrookes Cognitive Examination
- ☐ Abbreviated Mental Test Score
- ☐ Cambridge Examination for Mental Disorders of the Elderly

Your answer is incorrect.

Explanation:

Formerly known as the Dementia Questionnaire for Mentally Retarded Persons (DMR), The Dementia Questionnaire for People with Learning Disabilities (DLD) is used to facilitate the diagnosis of dementia in persons with intellectual disability, based on observations of caregivers.

Ref: Evenhuis HM. The dementia questionnaire for people with learning disabilities. In Neuropsychological assessments of dementia in down syndrome and intellectual disabilities 2018. pp. 43-56.

The correct answer is: Dementia Questionnaire for People with Learning Disabilities

Question 15

Not answered

Marked out of 1

Which of the following drugs is licensed for the treatment of self-injurious behaviour and aggression in patients with intellectual disability?

Select one:

- ☐ Lithium
- ☐ Fluoxetine
- ☐ Naltrexone
- ☐ Buspirone
- ☐ Carbamazepine

Your answer is incorrect.

Explanation:

Lithium is licensed for the treatment of self-injurious behaviour and aggression in individuals with intellectual disability (ID). Lithium is perhaps best considered when there is a sub-syndromal or nonspecific 'affective component'. Some authorities suggest that challenging behaviour may occur in the context of very rapid cycling bipolar disorder in some individuals with severe and profound ID.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 800.

The correct answer is: Lithium

Question 16

Not answered

Marked out of 1

Which of the following is correct concerning fragile X syndrome and IQ in males?

Select one:

- ☐ Adaptive behaviour declines with age
- ☐ Length of triplet repeats is inversely related to the IQ
- ☐ Abstract reasoning abilities are higher than expected during adolescence
- ☐ IQ stabilises around 20-30 yrs of age
- ☐ Cognitive degeneration is characteristic of the disease

Your answer is incorrect.

The higher the triplets, the lower the IQ. A drop in IQ is noted especially in males with Fragile X. This is most pronounced between ages 8-12 and stabilises by puberty (age 15). No further drop occurs after this, and, in fact, the adaptive skills improve continuously in most males. Hence, FXMR cannot be classed as a degenerative disease. The IQ drop phenomenon may be related to higher stress upon abstract abilities in different IQ tests used for adults and adolescents compared to children. Being concrete thinkers, most FXMR males show a drop in IQ. This drop is correlated with methylation status of FMR1 gene.

The correct answer is: Length of triplet repeats is inversely related to the IQ

Question 17

Not answered

Marked out of 1

Which of the following is not a feature of Cornelia de Lange syndrome?

Select one:

- ☐ Large hands and feet
- ☐ Microcephaly
- ☐ Hypertrichosis
- ☐ Continuous eyebrows
- ☐ Severe to profound intellectual disability

Your answer is incorrect.

Explanation:

Cornelia de Lange syndrome is a rare genetic disorder associated with a lack of pregnancy-associated plasma protein (PAPPA) linked to chromosome 9q33. A similar phenotype is associated with trisomy 5p, ring chromosome 3. Individuals with Cornelia de Lange syndrome have severe to profound intellectual disability. Clinical features include short stature, hypertrichosis (hirsutism, synophrys (continuous eyebrows), long eyelashes), prominent philtrum, small upturned nose, thin down-turning upper lip, microcephaly, low-set ears, and small hands and feet. It is also associated with GI problems (in particular reflux), congenital heart defects, visual and hearing problems, skin problems, epilepsy, and death in infancy. Behavioural problems include self-injury, temper tantrums, stereotypic movements, and twirling.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 819.

The correct answer is: Large hands and feet

Question 18

Not answered

Marked out of 1

Anxiety is a common association in the following disorders except

Select one:

- ☐ Williams syndrome
- ☐ Fragile X syndrome
- ☐ Down syndrome
- ☐ Prader-Willi syndrome
- ☐ Cornelia De Lange syndrome

Your answer is incorrect.

Some genetic causes of LD have associations with anxiety. For example, Fragile X syndrome is associated with social anxiety disorder, Prader-Willi syndrome and Rubinstein-Taybi are associated with OCD (Levitas and Read 1998), Williams syndrome is associated with anxiety disorder and phobias (Einfield et al. 2001) while Cornelia De Lange syndrome is associated with compulsive behaviour (Hyman et al. 2001).

The correct answer is: Down syndrome

Question 19

Not answered

Marked out of 1

Self-talk seems to be particularly common in children with

Select one:

- ☐ Prader-Willi syndrome
- ☐ Attention deficit hyperactivity disorder
- ☐ Depression
- ☐ Fragile X syndrome
- ☐ Down syndrome

Your answer is incorrect.

Explanation:

The behavioural phenotype of Down syndrome includes being passive, affable, demonstrating hyperactivity in childhood, being stubborn, and often talking or mumbling to themselves (private speech).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

Dykens EM et al. Psychiatric disorders in adolescents and young adults with Down syndrome and other intellectual disabilities. Journal of neurodevelopmental disorders. 2015 Dec;7(1):1-8.

The correct answer is: Down syndrome

Question 20

Not answered

Marked out of 1

What percentage of persons with intellectual disabilities have severe intellectual disability?

Select one:

- ☐ 1-2%
- ☐ 0.1%
- ☐ 12%
- ☐ 4%
- ☐ 0.85%

Your answer is incorrect.

Explanation:

Mild intellectual disability (ID) represents approximately 85% of persons with intellectual disabilities; moderate ID 10%; severe ID 4%; and profound ID 1-2%.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 95.

The correct answer is: 4%

Question **21**

Not answered

Marked out of 1

The prevalence of epilepsy among patients with a learning disability is about

Select one:

- ☐ 20-25%
- ☐ 5-10%
- ☐ 1-2%
- ☐ 50%
- ☐ 10-15%

Your answer is incorrect.

Epilepsy is commonly associated with numerous causes of LD e.g. Down's syndrome (5-10%), fragile X (25%), Angelman syndrome (90%), Rett syndrome (90%).

The correct answer is: 20-25%



Question 22

Not answered

Marked out of 1

Which of the following features is NOT suggestive of Lennox-Gastaut syndrome?

Select one:

- ☐ Multiple seizures types
- ☐ Diffuse interictal slow spike-waves on waking EEG
- ☐ Absence seizures and drop attacks
- ☐ Most often appears between ages of 3 and 5
- ☐ Progressive increase in IQ

Your answer is incorrect.

Explanation:

Lennox-Gastaut syndrome (LGS) is a severe form of epilepsy with onset in childhood (age of onset usually before 8 years; peak between 3 and 5 years). The syndrome is defined by a triad of multiple drug-resistant seizure types (tonic, atonic, and atypical absences); a specific EEG pattern showing burst of slow spike-wave complexes or generalised paroxysmal fast activity; and intellectual disability. The interictal EEG pattern of slow spike-waves is associated with LGS, though not pathognomonic of the disorder. Atonic seizures are common in LGS; they are also known as 'drop attacks' or 'drop seizures' because they are characterised by a sudden loss of muscle tone that can result in the patient falling to the ground. LGS has deleterious effects on intellectual functioning. Cognitive impairment usually becomes more apparent over time, and within 5 years of onset, serious intellectual problems have developed in most patients.

Ref: Asadi-Pooya AA. Lennox-Gastaut syndrome: a comprehensive review. Neurological Sciences. 2018 Mar 1;39(3):403-14.

Strzelczyk A, Schubert-Bast S. Expanding the treatment landscape for Lennox-Gastaut syndrome: current and future strategies. CNS drugs. 2021 Jan;35(1):61-83.

The correct answer is: Progressive increase in IQ

Question 23

Not answered

Marked out of 1

A male patient has autistic traits, conversational rigidity, palilalia, anxiety, and poor eye contact. Which of the following is the most likely diagnosis?

Select one:

- ☐ Lesch Nyhan syndrome
- ☐ Down syndrome
- ☐ Turner syndrome
- ☐ Klinefelter syndrome
- ☐ Fragile X syndrome

Your answer is incorrect.

Explanation:

Fragile X syndrome is the most common inherited cause of intellectual disability and the most common single gene cause of autism. Behavioural features of fragile X syndrome appear to be similar to those seen in ADHD and autism, including hand flapping/waving, repetitive mannerisms, shyness, gaze avoidance, poor peer relationships, communication difficulties (e.g. delayed language development, conversational rigidity, perseveration, echolalia, palilalia, cluttering, and over-detailed/circumstantial speech), and psychiatric problems (e.g. prominent depression/anxiety).

Ref: Razak KA et al. Developmental studies in fragile X syndrome. Journal of neurodevelopmental disorders. 2020 Dec;12(1):1-5.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 814.

The correct answer is: Fragile X syndrome

Question **24**

Not answered

Marked out of 1

Which of the following sleep patterns is commonly seen in Rett syndrome?

Select one:

- ☐ Lack of daytime sleep
- ☐ Excessive nocturnal sleep
- ☐ Drop attacks and sleep paralysis
- ☐ Excessive somnolence
- ☐ Sleepwalking

Your answer is incorrect.

Explanation:

Sleep disturbances are relatively common in Rett syndrome. Typically, patients suffer from initial and/or intermediate insomnia (i.e., difficulty falling asleep and frequent awakenings during the night). Complaints of excessive somnolence, and of initiating and maintaining sleep are most prevalent in Rett syndrome.

Ref: Persico AM et al. The psychopharmacology of autism spectrum disorder and Rett syndrome. In Handbook of clinical neurology 2019 Jan 1 (Vol. 165, pp. 391-414). Elsevier.

Spruyt K. Sleep problems in individuals with Rett Syndrome: A Systematic Review and Meta-analysis. Sleep Epidemiology. 2022 Apr 14:1-12.

The correct answer is: Excessive somnolence

Question 25

Not answered

Marked out of 1

Which of the following suggests a 'subcultural' rather than a biological cause for intellectual disability?

Select one:

- ☐ Equal distribution across families of different types of socio-economic status
- ☐ Intellectual ability is impaired in family members
- ☐ Dysmorphic features are common
- ☐ Significant impairment in intellectual abilities
- ☐ Males are more commonly affected than females

Your answer is incorrect.

Explanation:

Although the concept of 'psychosocial' causation (due to physical and emotional neglect) for intellectual disability is controversial, mild or borderline intellectual impairment is more common in families of lower socio-economic status (SES). This is best viewed as a cultural norm, and individuals generally have no, or only minor, impairments in adaptive functioning. Generally, the intellectual ability of family members is also in the borderline range, dysmorphic characteristics are less likely, and other impairments or disabilities are unusual ('subcultural' intellectual disability). This is in contrast to biological causation where impairments are more significant, there is no difference in SES, parents and siblings are usually of normal intelligence, and dysmorphic features are more common.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 791.

The correct answer is: Intellectual ability is impaired in family members

Question 26

Not answered

Marked out of 1

Which of the following is the most common aetiological factor in developing intellectual disability?

Select one:

- ☐ Defects in embryogenesis
- ☐ Acquired nutritional defects
- ☐ Obstetric complications
- ☐ Genetic loading in families
- ☐ Exposure to environmental toxins

Your answer is incorrect.

Explanation:

Aetiological contributions to intellectual disability may be classified based on the timing of the causative event: 50-70% of cases attributable to a prenatal factor (during embryogenesis), 10-20% to a perinatal factor (around the time of delivery), and 5-10% to a postnatal factor (in the first years of life).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 794-795.

The correct answer is: Defects in embryogenesis

Question 27

Not answered

Marked out of 1

Among people with intellectual disability, pica is more commonly seen in which of the following conditions?

Select one:

- ☐ Rett syndrome
- ☐ Autism spectrum disorder
- ☐ Fragile X syndrome
- ☐ Down syndrome
- ☐ Foetal alcohol syndrome

Your answer is incorrect.

Explanation:

Pica is characterised by persistent (>1 month) eating of non-nutritive substances at a developmentally inappropriate age (>1 year). It typically occurs during the 2nd and 3rd years of life, though may occur in young pregnant women. Common substances include dirt, stones, plastic, paper, hair, faeces, wood, string, etc. Pica is more frequent in people with intellectual disability and autism spectrum disorder.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 694.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 162-163.

The correct answer is: Autism spectrum disorder

Question 28

Not answered

Marked out of 1

Which of the following techniques is used to treat stuttering?

Select one:

- ☐ Forced speech
- ☐ Makaton speech
- ☐ Tactical speech
- ☐ Delayed speech
- ☐ Prolonged speech

Your answer is incorrect.

Explanation:

Treatment of stuttering falls into two general categories: indirect and direct. Indirect treatment does not use overt or explicit methods with the child. Rather, it targets parent counselling, and/or the use of the parent/clinician to model fluency facilitators for the child. Direct treatments usually explicitly teach the child to change something about the way the child speaks. Direct treatments may focus on obtaining 'smooth speech' through fluency shaping strategies based on prolonged speech principles that include gradual phonatory initiation, smooth transitions between words using imagery that appeals to children, or stuttering modification. The clinician and/or parent models a slow, relaxed speaking pattern that emphasises reduced rate, smooth articulatory transitions, and slightly prolonged consonants and vowels, while preserving the natural intonation and stress patterns.

Shenker RC, Santayana G. What are the options for the treatment of stuttering in preschool children? In Seminars in speech and language 2018 Sep (Vol. 39, No. 04). pp. 313-323.

The correct answer is: Prolonged speech

Question 29

Not answered

Marked out of 1

Which of the following conditions is associated with skin picking?

Select one:

- ☐ Cornelia de Lange syndrome
- ☐ Angelman syndrome
- ☐ Prader-Willi syndrome
- ☐ Down syndrome
- ☐ Fragile X syndrome

Your answer is incorrect.

Explanation:

The behavioural phenotype of Prader-Willi syndrome includes compulsive behaviour, hyperphagia, hoarding, impulsivity, emotional lability, tantrums, excessive daytime sleepiness, skin-picking, and aggression.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

The correct answer is: Prader-Willi syndrome

Question 30

Not answered

Marked out of 1

A 43-year-old elderly primi was worried about having a baby with Down syndrome has come to see you for advice. What is the chance of her delivering a child with Down syndrome?

Select one:

- ☐ 1 in 20 chance
- ☐ 1 in 1500 chance
- ☐ 1 in 10 chance
- ☐ 1 in 50 chance
- ☐ 1 in 100 chance

Your answer is incorrect.

The overall incidence of Down syndrome is approximately 1 in 700 live births, although a further 50% result in a spontaneous abortion and 20% are stillborn. Incidence increases with maternal age, so at 20 years of age a mother has a 1 in 1500 chance of delivering a child with downs syndrome but this is increased to a 1 in 50 chance at 43 years of age. Ref; Core Psychiatry: ed. Wright, Stern and Phelan Pg 162

The correct answer is: 1 in 50 chance

Question **31**

Not answered

Marked out of 1

You are counselling a mother who has one child with autism and is expecting her second. She wishes to know the risk of autism in this second child. The correct answer is

Select one:

- ☐ 1-10%
- ☐ 40 - 50%
- ☐ 30-40%
- ☐ 10 - 20%
- ☐ 50 - 60%

Your answer is incorrect.

The risk of autism in a sibling of an autistic child is estimated to be less than 10%, provided only one child in the family has autism so far. If more than one child has autism, the next (expected) child has nearly 35% risk.

Szatmari reported sibling risk for autism to range from 2% to 6% (average 2.2%). When other PDD diagnoses are included, the sibling risk increases to 5.5%. The risk of autism for second-degree relatives is 0.18%, and 0.12% for third-degree relatives. 70% of the monozygotic (MZ) twins were concordant for autism, whereas none of the dizygotic (DZ) twins were concordant in a small UK sample. When autistic spectrum is broadly defined, concordance increased to 87% for MZ and to 9% for DZ twins. Szattmari P, Jones MB. Genetic epidemiology of autism and pervasive developmental disorders. In: FR Volkmar (ed), Cambridge University Press, 2006.

The correct answer is: 1-10%

Question **32**

Not answered

Marked out of 1

Which of the following is a sex chromosome monosomy?

Select one:

- ☐ Klinefelter syndrome
- ☐ Fragile X syndrome
- ☐ XYY male
- ☐ Trisomy X
- ☐ Turner syndrome

Your answer is incorrect.

Explanation:

Turner syndrome is a sex chromosome monosomy with karyotype 45,XO and female phenotype.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 817.

The correct answer is: Turner syndrome



Question **33**

Not answered

Marked out of 1

Which of the following chromosomal abnormalities is most likely to cause intellectual disability?

Select one:

- ☐ Fusion of chromosomes 21 and 15
- ☐ XXXY
- ☐ XXY
- ☐ Extra chromosome 21
- ☐ XO

Your answer is incorrect.

Explanation:

Down syndrome is a trisomy of chromosome 21 and the most common genetic cause of intellectual disability (ID). IQ most often below 50 (range: low to high/moderate ID).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 806.

The correct answer is: Extra chromosome 21

Question **34**

Not answered

Marked out of 1

The behavioural phenotype of Prader-Willi syndrome commonly includes which of the following?

Select one:

- ☐ Extraordinary verbal facility
- ☐ Gaze aversion and social avoidance
- ☐ Extreme self-mutilation
- ☐ Skin picking
- ☐ Severe sleep disturbance

Your answer is incorrect.

Explanation:

The behavioural phenotype of Prader-Willi syndrome includes compulsive behaviour, hyperphagia, hoarding, impulsivity, emotional lability, tantrums, excessive daytime sleepiness, skin-picking, and aggression.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

The correct answer is: Skin picking

Question 35

Not answered

Marked out of 1

Which of the following is a postnatal factor that increases the risk of intellectual disability?

Select one:

- ☐ Neonatal septicaemia
- ☐ Gestational diabetes in the mother
- ☐ Birth asphyxia
- ☐ Genetic mutations
- ☐ Head injury

Your answer is incorrect.

Explanation:

Postnatal factors (occurring in the first years of life) associated with intellectual disability include CNS infections, intracranial tumours, hypoxic brain injury, head injury (e.g. RTAs, household accidents, child abuse), exposure to toxic agents, and the psychosocial environment (i.e. deprivation).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 795.

The correct answer is: Head injury

Question 36

Not answered

Marked out of 1

Those with Down syndrome who survive into their 40s and 50s show pathological brain changes similar to

Select one:

- ☐ Frontotemporal dementia
- ☐ Alzheimer dementia
- ☐ Parkinson disease
- ☐ Huntington disease
- ☐ Vascular dementia

Your answer is incorrect.

Explanation:

Neuropathological changes similar to Alzheimer dementia are seen in individuals with Down syndrome over the age of 40. Other CNS abnormalities in Down syndrome include reduced brain weight, reduced gyri, cortical thinning, underdeveloped middle lobe of the cerebellum, reduced neuronal numbers in the cerebellum/locus caeruleus/basal forebrain, reduced cholinergic neurons, and epilepsy.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 807.

The correct answer is: Alzheimer dementia

Question **37**

Not answered

Marked out of 1

Lesch-Nyhan syndrome is characterised by

Select one:

- ☐ Hydrocephalus
- ☐ Normal secondary sexual characters
- ☐ Hepatic failure
- ☐ Compulsive self-mutilation
- ☐ Hypertonia

Your answer is incorrect.

Explanation:

Lesch-Nyhan syndrome is an extremely rare recessive condition associated with a defect on the short arm of the X chromosome. A deficiency of hypoxanthine phosphoribosyl transferase leads to build up of uric acid. Severe compulsive self-mutilation is often seen, including biting of the lips, inside of the mouth, and fingers.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 816.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

The correct answer is: Compulsive self-mutilation

Question **38**

Not answered

Marked out of 1

In which of the following conditions is microcephaly a characteristic feature?

Select one:

- ☐ Fragile X syndrome
- ☐ Down syndrome
- ☐ Foetal alcohol syndrome
- ☐ Tuberous sclerosis
- ☐ Hydrocephalus

Your answer is incorrect.

Explanation:

Clinical features of foetal alcohol syndrome include microcephaly, short palpebral fissures, small maxillae and mandibles, flat philtrum, thin upper lip, and short stature. The behavioural phenotype includes mild to moderate intellectual disability, irritability, inattention, and memory impairment.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 818.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 87.

The correct answer is: Foetal alcohol syndrome

Question 39

Not answered

Marked out of 1

The most common inherited cause of intellectual disability is

Select one:

- ☐ Down syndrome
- ☐ Hypothyroidism
- ☐ Fragile X syndrome
- ☐ Neurofibromatosis
- ☐ Cri du chat syndrome

Your answer is incorrect.

Explanation:

Fragile X syndrome is the most common inherited cause of intellectual disability and the most common single gene cause of autism. The syndrome affects around 1/4000 males and 1/8000 females.

Ref: Razak KA et al. Developmental studies in fragile X syndrome. Journal of neurodevelopmental disorders. 2020 Dec;12(1):1-5.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 814.

The correct answer is: Fragile X syndrome

Question 40

Not answered

Marked out of 1

All of the following are true when comparing high-functioning autism (HFA) with Asperger's Disorder except

Select one:

- ☐ IQ levels of the two groups are comparable
- ☐ Differences in language between these groups reduce with age
- ☐ Language onset is delayed for HFA but not Asperger's
- ☐ High functioning autism is more common in girls while Asperger's is more common in boys
- ☐ There is little evidence to support the retention of two diagnostic categories

Your answer is incorrect.

Both HFA and Asperger's syndrome are more common in boys. Macintosh and Dissanayake (2004) noted that although different language patterns were evident (the HFA group demonstrating more atypical speech patterns [later speech onset, echolalia, noun reversals, atypical gestures], while the Asperger 's group tended to have more verbal rituals and asked odd questions), studies demonstrated little basis for differentiation on the basis of diagnosis. Although the HFA group demonstrated greater dysfunction in preschool, by adolescence, the only distinguishing feature between these two groups was that the teens with Asperger 's disorder engaged in more pedantic dialogues.

The correct answer is: High functioning autism is more common in girls while Asperger's is more common in boys

Question 41

Not answered

Marked out of 1

Which of the following statements is correct about velo-cardio-facial syndrome?

Select one:

- ☐ Large ears and wide palpebral fissures are common features
- ☐ It is usually associated with severe intellectual disability
- ☐ It occurs in around 1 in 2000 people
- ☐ It is due to a chromosome 21 microdeletion
- ☐ It is usually associated with dementia

Your answer is incorrect.

Explanation:

Velo-cardio-facial (di George) syndrome is caused by a microdeletion of chromosome 22 (22q11.2). The incidence is 1:2000. Clinical features include intellectual disability (mild: two-thirds; moderate: one-third), cardiac abnormalities (75%), facial features (microcephaly, cleft palate, small mouth, long face, narrow palpebral fissure, minor ear abnormalities), hypocalcemia (60%), hypospadias, and long & thin hands. Associated behavioural and psychiatric disorders include schizophrenia and blunted/inappropriate affect.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 808.

The correct answer is: It occurs in around 1 in 2000 people

Question 42

Not answered

Marked out of 1

Which of the following conditions is most likely to be associated with hypertrichosis and severe intellectual disability?

Select one:

- ☐ Klinefelter syndrome
- ☐ Neurofibromatosis
- ☐ Lesch-Nyhan syndrome
- ☐ Prader-Willi syndrome
- ☐ Cornelia de Lange syndrome

Your answer is incorrect.

Explanation:

Cornelia de Lange syndrome is a rare genetic disorder associated with a lack of pregnancy-associated plasma protein (PAPPA) linked to chromosome 9q33. A similar phenotype is associated with trisomy 5p, ring chromosome 3. Individuals with Cornelia de Lange syndrome have severe to profound intellectual disability. Clinical features include short stature, hypertrichosis (hirsutism, synophrys (continuous eyebrows), long eyelashes), prominent philtrum, small upturned nose, thin down-turning upper lip, microcephaly, low-set ears, and small hands and feet. It is also associated with GI problems (in particular reflux), congenital heart defects, visual and hearing problems, skin problems, epilepsy, and death in infancy. Behavioural problems include self-injury, temper tantrums, stereotypic movements, and twirling.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 819.

The correct answer is: Cornelia de Lange syndrome

Question 43

Not answered

Marked out of 1

Which of the following is inherited in an autosomal dominant manner?

Select one:

- ☐ Sanfillipo disease
- ☐ Hunter syndrome
- ☐ Tuberous sclerosis
- ☐ Hurler syndrome
- ☐ Lesch-Nyhan syndrome

Your answer is incorrect.

Explanation:

Tuberous sclerosis is an autosomal dominant syndrome. The incidence is equal between males and females. Individuals show varying degrees (usually severe) of intellectual disability (50%).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 86.

The correct answer is: Tuberous sclerosis

Question 44

Not answered

Marked out of 1

Prominent anxiety is most common in which of the following syndromes associated with intellectual disability?

Select one:

- ☐ Foetal alcohol syndrome
- ☐ Fragile X syndrome
- ☐ Down syndrome
- ☐ Congenital hypothyroidism
- ☐ Lesch-Nyhan syndrome

Your answer is incorrect.

Explanation:

Fragile X syndrome is the most common inherited cause of intellectual disability and the most common single gene cause of autism. Behavioural features of fragile X syndrome appear to be similar to those seen in ADHD and autism, including hand flapping/waving, repetitive mannerisms, shyness, gaze avoidance, poor peer relationships, communication difficulties (e.g. delayed language development, conversational rigidity, perseveration, echolalia, palilalia, cluttering, and over-detailed/circumstantial speech), and psychiatric problems (e.g. prominent depression/anxiety).

Ref: Razak KA et al. Developmental studies in fragile X syndrome. Journal of neurodevelopmental disorders. 2020 Dec;12(1):1-5.

Sample D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 814.

The correct answer is: Fragile X syndrome

Question 45

Not answered

Marked out of 1

Happy disposition and paroxysmal laughter are features commonly seen in

Select one:

- ☐ Lesch-Nyhan syndrome
- ☐ Angelman syndrome
- ☐ Cornelia de Lange syndrome
- ☐ Prader-Willi syndrome
- ☐ Williams syndrome

Your answer is incorrect.

Explanation:

Angelman ('happy puppet') syndrome is characterised by severe/profound intellectual disability. Approximately 86% of individuals have epilepsy. Other clinical features include happy disposition, paroxysmal laughter, absence of speech, hand flapping, clapping, ataxia (jerky limb movements, gait problems), and anecdotal love of water and music.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 808.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

The correct answer is: Angelman syndrome

Question 46

Not answered

Marked out of 1

A microdeletion on the paternal 15q12 chromosomal segment causes

Select one:

- ☐ Prader-Willi syndrome
- ☐ Angelman syndrome
- ☐ Lesch-Nyhan syndrome
- ☐ Williams syndrome
- ☐ Cornelia de Lange syndrome

Your answer is incorrect.

Explanation:

Prader-Willi syndrome is a rare genetic disorder associated with a microdeletion on chromosome 15 (q11-13), usually on the paternal chromosome. 90% of cases are sporadic. The candidate gene is small nuclear ribonucleoprotein polypeptide (SNRPN). Males are more commonly affected (M:F = 4:3) and the incidence is 1:10,000-40,000.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 808.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

The correct answer is: Prader-Willi syndrome

Question 47

Not answered

Marked out of 1

Predisposing factors for behavioural problems in children with intellectual disability include

Select one:

- ☐ Poor communication
- ☐ Limited coping strategies
- ☐ Sensory impairments
- ☐ Epilepsy
- ☐ All of the listed options

Your answer is incorrect.

Explanation:

Factors associated with behavioural disorders and challenging behaviour in intellectual disability include severe intellectual impairment, poor/absent language ability, poor social comprehension, high emotionality, hyperactivity, poor sociability, physical problems (e.g. epilepsy, cerebral palsy, cardiac problems, GI problems), visual/hearing impairment, and communication difficulties.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 828-829.

The correct answer is: All of the listed options

Question **48**

Not answered

Marked out of 1

Approximately 20% of those with autism spectrum disorders also have

Select one:

- ☐ Obesity
- ☐ Language impairment
- ☐ Polydactyly
- ☐ Intellectual disability
- ☐ Macrocephaly

Your answer is incorrect.

Explanation:

Autism spectrum disorders (ASD) increase the risk of obesity. The prevalence of obesity in ASD is approximately 20-22%.

Ref: Kahathuduwa CN et al. The risk of overweight and obesity in children with autism spectrum disorders: A systematic review and meta-analysis. Obesity Reviews. 2019 Dec;20(12):1667-79.

The correct answer is: Obesity



Question 49

Not answered

Marked out of 1

Klinefelter syndrome is characterised by all of the following except

Select one:

- ☐ Hypogonadism
- ☐ Taller than average stature
- ☐ Median IQ of 90
- ☐ Karyotype 47XXY
- ☐ Higher incidence in prison populations

Your answer is incorrect.

Explanation:

Klinefelter syndrome (karyotype 47,XXY) is one of the most common chromosomal disorders. It is seen in 1:1000 live male births. Clinical features include a variable degree of development of secondary sexual characteristics (with sparse body hair, hypogonadism, and gynaecomastia), taller than average stature (by around 4cms), thin build, and median IQ of approximately 90 with skewed distribution (most in 60-70 range). There is an uncertain association with psychiatric disorders.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 817.

The correct answer is: Higher incidence in prison populations

Question 50

Not answered

Marked out of 1

What is the mode of inheritance of Fragile X syndrome?

Select one:

- ☐ Autosomal dominant
- ☐ Mitochondrial gene transmission
- ☐ X linked recessive
- ☐ Autosomal recessive
- ☐ X-linked dominant

Your answer is incorrect.

Explanation:

Fragile X syndrome is a common inherited cause of intellectual disability (ID) with X-linked dominant transmission. Penetrance is low, but greater in males than females due to the 'protective' effects of the second normal X chromosome in females. Fragile X syndrome accounts for 10-12% of ID in men.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 814.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

The correct answer is: X-linked dominant

Question 51

Not answered

Marked out of 1

Which of the following is true about the epidemiology of autism?

Select one:

- ☐ It is more common in children who have had the MMR vaccine
- ☐ The heritability is approximately 90%
- ☐ It is more common in girls than boys
- ☐ It is more common in parents of higher socio-economic status
- ☐ Monozygotic twin concordance is greater than 90%

Your answer is incorrect.

Explanation:

Twin and family studies consistently demonstrate that autism has a particularly large genetic contribution, with a heritability of approximately 80-90%. Autism is highly heritable and is among the most heritable common medical conditions.

Ref: Lord C et al. Autism spectrum disorder. Nature reviews Disease primers. 2020 Jan 16;6(1):1-23.

Sandin S et al. The heritability of autism spectrum disorder. Jama. 2017 Sep 26;318(12):1182-4.

The correct answer is: The heritability is approximately 90%

Question 52

Not answered

Marked out of 1

In individuals with intellectual disability, self-injurious behaviour occurs most frequently between the ages of

Select one:

- ☐ 5 and 20 years
- ☐ 1 and 2 years
- ☐ 25 and 50 years
- ☐ 2 and 3 years
- ☐ More than 50 years

Your answer is incorrect.

Explanation:

Many individuals with intellectual disability suffer from self-injurious behaviour, which includes head-banging, head hitting, and other behaviours that can cause physical trauma. Self-injurious behaviour in those with intellectual disability often emerges between 2-3 years of age. It is more frequent and severe in childhood and adolescence. The severity can decrease over the course of adulthood.

Ref: Hagopian LP, Leoni M. Self-injurious behavior among individuals with intellectual and developmental disabilities. Acta Psychopathol. 2017;3(5):70.

The correct answer is: 5 and 20 years

Question 53

Not answered

Marked out of 1

Which of the following is the most common chromosomal aberration seen in Down syndrome?

Select one:

- ☐ Mosaicism with both normal and trisomic cell lines
- ☐ Translocation between chromosome 21 and 17
- ☐ Full trisomy 21
- ☐ Translocation between chromosome 21 and 15
- ☐ Mosaicism with only trisomic cells present

Your answer is incorrect.

Explanation:

Down syndrome is the most common genetic cause of intellectual disability. Full trisomy 21 (nondisjunction) occurs in around 95% of cases. Extra chromosome 21 material can also occur due to Robertsonian translocation in 5% cases (long arm of chromosome 21 is attached to another chromosome, usually chromosome 14; also 13, 15, and 22 described). In 2-5% cases, there is a mixture of normal and trisomic cell lines, known as mosaic Down syndrome.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 806.

The correct answer is: Full trisomy 21

Question **54**

Not answered

Marked out of 1

Which of the following is an X-linked recessive condition?

Select one:

- ☐ Hurler syndrome
- ☐ Tuberous sclerosis
- ☐ Hunter syndrome
- ☐ Sanfillipo disease
- ☐ Niemann-Pick disease

Your answer is incorrect.

Explanation:

Hunter syndrome is an X-linked recessive syndrome caused by a deficiency of iduronate sulfatase (mapped to Xq27-28) and consequent accumulation of glycosaminoglycans. The incidence is 1:100,000.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 816.

The correct answer is: Hunter syndrome

Question 55

Not answered

Marked out of 1

Which of the following is most likely to be true while diagnosing PTSD in patients with learning disability?

Select one:

- ☐ Nightmares are common
- ☐ Startle response is common
- ☐ Aggression may be the most common presenting problem.
- ☐ Never needs a history of trauma
- ☐ Remission of symptoms is often within few days

Your answer is incorrect.

In people with learning disabilities, aggression is the most common presenting symptom of post-traumatic stress disorder, as opposed to sleep problems, nightmares and 'jumpiness', which are commonly reported in the non-disabled population. Doyle & Mitchell (2003) reported that 16.5% of psychiatric referrals to paediatric learning disability teams meet the criteria for post-traumatic stress disorder. Doyle and Mitchell, Journal of Intellectual Disabilities 2003; 7; 23

The correct answer is: Aggression may be the most common presenting problem.

Question 56

Not answered

Marked out of 1

Which of the following statements is true regarding posttraumatic stress disorder in adults with learning disabilities?

Select one:

- ☐ Nightmares are seldom reported
- ☐ An exaggerated startle response is often seen
- ☐ Co-morbid depressive symptoms are less likely than in adults with average intellect
- ☐ Symptoms resolve within a few weeks following the traumatic event
- ☐ It is highly likely to be diagnosed in adults with severe learning disabilities

Your answer is incorrect.

In studies of PTSD in the general adult population with learning disability, the symptoms most commonly reported were nightmares, a feeling of jumpiness and trouble sleeping (Helzer et al. 1987). In half of those with symptoms, the symptoms lasted less than 6 months, but for one-third the symptoms persisted for over a year. It is unusual for PTSD to occur in isolation and comorbidity with other psychiatric disorders is the norm. People with learning disability are at greater risk for the psychiatric disorder than the general population (Rutter et al 1970). Individuals with mild to profound learning disability suffer disorders similar to those affecting the general population. The main difficulty is the diagnosis, as the presentations of a disorder may be different in those with severe developmental delay. Therefore, the diagnosis of PTSD in those who have a severe learning disability, who do not have the communication skills to describe their thoughts, feelings and mood may be hindered. At times we are reliant on observing changes in behaviour and function, for example, aggression, regressive behaviours and changes in sleep pattern. Owing to the limited recognition of PTSD in people with learning disability these symptoms may be attributed to other psychiatric diagnoses. (Excerpt was retrieved from McCarthy J (2001), PTSD in learning disability <http://apt.rcpsych.org/content/aptrcpsych/7/3/163.full.pdf>)

The correct answer is: An exaggerated startle response is often seen

Question **57**

Not answered

Marked out of 1

A 20-year-old man with cerebral palsy and chronic epilepsy resides in a care home. His communication skills are very limited. He has limited ability to self-care, cannot contribute to self-maintenance despite complete supervision, and requires 24-hour supervised care. The severity of his intellectual disability is best classified as

Select one:

- ☐ Moderate
- ☐ Mild
- ☐ Profound
- ☐ Severe
- ☐ Borderline

Your answer is incorrect.

Explanation:



Level of intellectual disability	Level of functioning
Mild	Often struggle with academic skills and complex language comprehension Most are capable of basic self-care and domestic and practical activities Most are able to live relatively independently Capable of finding employment as adults, but may need appropriate support
Moderate	Capacity for academic skills and language comprehension varies, usually limited to basic skills Some can manage basic self-care and domestic and practical activities Most need significant ongoing support to live independently and maintain employment as adults
Severe	Very limited capacity for language and ability to gain academic skills Often accompanied by motor impairments Daily support in a supervised environment is usually required Some may develop basic self-care skills with intensive support and training
Profound	Very limited communication ability Potential to gain academic skills is limited to basic concrete skills Motor and sensory impairments usually present Daily support in a supervised care environment usually required

Adapted from Williams et al. 2018

SPMM COURSE



Ref: Williams F et al. Alcohol use disorders in people with intellectual disability. BJPsych Advances. 2018 Jul;24(4):264-72.

The correct answer is: Profound



Question 58

Not answered

Marked out of 1

The IQ of an 11-year-old boy is evaluated as 75. Which of the following is true?

Select one:

- ☐ Low IQ at age 11 is a definite predictor of conduct disorder at a later age
- ☐ There will be a further drop in IQ after adolescence
- ☐ There is an increased risk of schizophrenia compared to general population
- ☐ He falls two percentiles below the average IQ of the standard population
- ☐ None of the listed options

Your answer is incorrect.

Explanation:

Individuals with borderline intellectual function have an IQ of 71-84 (1-2 standard deviations below the mean on a normal curve). They are twice as likely to develop a psychiatric disorder, including schizophrenia, as people with average or above average IQ.

Ref: Hassiotis A et al. Association of Borderline Intellectual Functioning and Adverse Childhood Experience with adult psychiatric morbidity. Findings from a British birth cohort. BMC psychiatry. 2019 Dec;19(1):1-9.

The correct answer is: There is an increased risk of schizophrenia compared to general population

Question 59

Not answered

Marked out of 1

Of the following, the risk of developing Alzheimer disease in Down syndrome is most increased by

Select one:

- ☐ Obesity
- ☐ Chronic anxiety
- ☐ A family history of Alzheimer disease
- ☐ Mild intellectual disability rather than severe intellectual disability
- ☐ Eczema

Your answer is incorrect.

Explanation:

Almost all adults with Down syndrome (DS) develop Alzheimer disease (AD)-like neuropathology by 40 years of age. AD is characterised by a long asymptomatic preclinical stage in which amyloid pathology develops approximately 15-20 years before any cognitive impairment is observed. This preclinical stage is present in the autosomal dominant forms of AD and in early-onset AD associated with DS. The evidence is compelling that an increased dosage of *APP* (an HSA1 gene that encodes amyloid precursor protein) in trisomy 21 is necessary for an increased risk of AD in people with DS. The cognitive decline needs to be understood in the context of the existing cognitive phenotype and individual risk, including the APOE genotype, as the presence of the APOE E4 allele seems to increase risk of earlier cognitive decline due to AD.

Ref: Antonarakis SE et al. Down syndrome. Nature Reviews Disease Primers. 2020 Jan;6(1):9.

The correct answer is: A family history of Alzheimer disease

Question 60

Not answered

Marked out of 1

The highest incidence of intellectual disabilities is seen in

Select one:

- ☐ Infants
- ☐ Toddlers
- ☐ Early adulthood
- ☐ School-age children
- ☐ Older adults

Your answer is incorrect.

Explanation:

The prevalence of intellectual disability (ID) at any one time is estimated to range from 1-3% of the population. The incidence of ID is difficult to calculate because mild disabilities may be unrecognised until middle childhood. The highest incidence of ID is reported in school-age children, with a peak at ages 10-14 years. ID is about 1.5 times more common among males than females.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 100.

The correct answer is: School-age children

Question 61

Not answered

Marked out of 1

Clinical features of Prader-Willi syndrome include all except

Select one:

- ☐ Mild to moderate intellectual disability
- ☐ Orthopaedic deformities
- ☐ Macro-orchidism
- ☐ Overeating
- ☐ Small hands and feet

Your answer is incorrect.

Explanation:

Clinical features of Prader-Willi syndrome include hypotonia, failure to thrive in infancy, obesity, small hands and feet, micro-orchidism, cryptorchidism, short stature, almond-shaped eyes, fair hair and light skin, flat face, scoliosis, orthopaedic problems, mild to moderate intellectual disability, and hyperphagia.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

Sample D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 808.

The correct answer is: Macro-orchidism

Question 62

Not answered

Marked out of 1

Which of the following is true of offending in the intellectual disability (ID) population?

Select one:

- ☐ Moderate ID is strongly associated with homicide
- ☐ Conviction for arson leads to a fixed prison sentence
- ☐ Most offences are committed by those with borderline and mild ID
- ☐ People with ID commit the majority of violent offences
- ☐ Fire setting is the most commonly committed offence

Your answer is incorrect.

Explanation:

Offending occurs more often in people with milder form of intellectual disability (ID) than in those with severe ID. People with ID are at increased risk for crimes of aggression, sexual offending, and arson.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 747, 832.

The correct answer is: Most offences are committed by those with borderline and mild ID

Question 63

Not answered

Marked out of 1

Epilepsy is present in nearly 90 percent of individuals with

Select one:

- ☐ Angelman syndrome
- ☐ Fragile X syndrome
- ☐ Prader-Willi Syndrome
- ☐ Down syndrome
- ☐ Klinefelter syndrome

Your answer is incorrect.

Explanation:

Epilepsy is more common in people with intellectual disability (ID) than in the general population. The prevalence of epilepsy varies with the severity and cause of ID. It is higher in severe ID (30-50%) than in mild ID (15-20%). Epilepsy is common in people with ID of various causes, such as Down syndrome 5-10%; Fragile X syndrome 25%; Angelman syndrome 90%; and Rett syndrome 90%.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 824.

The correct answer is: Angelman syndrome

Question 64

Not answered

Marked out of 1

Which of the following conditions is most commonly associated with tall stature?

Select one:

- ☐ Angelman syndrome
- ☐ Turner syndrome
- ☐ Prader-Willi syndrome
- ☐ Klinefelter syndrome
- ☐ Cornelia de Lange syndrome

Your answer is incorrect.

Explanation:

Klinefelter syndrome (karyotype 47,XXY) is one of the most common chromosomal disorders. It is seen in 1:1000 live male births. Clinical features include a variable degree of development of secondary sexual characteristics (with sparse body hair, hypogonadism, and gynaecomastia), taller than average stature (by around 4cms), thin build, and median IQ of approximately 90 with skewed distribution (most in 60-70 range).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 817.

The correct answer is: Klinefelter syndrome

Question 65

Not answered

Marked out of 1

A 'subcultural' rather than a biological explanation for intellectual disability is supported by which of the following observations?

Select one:

- ☐ Normal distribution of intellectual disability across socio-economic groups
- ☐ Borderline IQ in parents and siblings
- ☐ Presence of facial dysmorphology
- ☐ Presence of moderate or severe intellectual disability
- ☐ Impairments in adaptive functioning

Your answer is incorrect.

Explanation:

Although the concept of 'psychosocial' causation (due to physical and emotional neglect) for intellectual disability is controversial, mild or borderline intellectual impairment is more common in families of lower socio-economic status (SES). This is best viewed as a cultural norm, and individuals generally have no, or only minor, impairments in adaptive functioning. Generally, the intellectual ability of family members is also in the borderline range, dysmorphic characteristics are less likely, and other impairments or disabilities are unusual ('subcultural' intellectual disability). This is in contrast to biological causation where impairments are more significant, there is no difference in SES, parents and siblings are usually of normal intelligence, and dysmorphic features are more common.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 791.

The correct answer is: Borderline IQ in parents and siblings

Question 66

Not answered

Marked out of 1

In adults with intellectual disability, dysphagia is a known side effect of which of the following drugs?

Select one:

- ☐ Propranolol
- ☐ Fluoxetine
- ☐ Naltrexone
- ☐ Clonazepam
- ☐ Lithium

Your answer is incorrect.

Explanation:

Medications that depress the central nervous system, including clonazepam and other benzodiazepines, can lead to a decrease in awareness and voluntary muscle control which may affect swallowing in people with intellectual disability.

Ref: Morton-Nance S. Drug-induced dysphagia and the high-risk effect on people with intellectual disabilities. Learning Disability Practice. 2020 Nov 26;23(6).

The correct answer is: Clonazepam

Question **67**

Not answered

Marked out of 1

The most common cause of death in Down syndrome is

Select one:

- ☐ Mitral valve prolapse
- ☐ Gastrointestinal anomalies
- ☐ Cardiac arrhythmias
- ☐ Pulmonary hypertension
- ☐ Infections

Your answer is incorrect.

Explanation:

Cardiovascular and pulmonary diseases account for 75% of mortality in Down syndrome. The leading causes of death are pneumonia and other infectious lung disease.

Ref: Colvin KL, Yeager ME. What people with Down Syndrome can teach us about cardiopulmonary disease. European Respiratory Review. 2017 Mar 31;26(143).

The correct answer is: Infections

Question 68

Not answered

Marked out of 1

Which of the following conditions is seen almost exclusively in girls?

Select one:

- ☐ Down syndrome
- ☐ Rett syndrome
- ☐ Tuberous sclerosis
- ☐ Angelman syndrome
- ☐ Rubenstein-Taybi syndrome

Your answer is incorrect.

Explanation:

Rett syndrome is believed to be caused by a dominant X-linked gene. It is a degenerative disorder that almost exclusively affects females.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 103.

Persico AM et al. The psychopharmacology of autism spectrum disorder and Rett syndrome. In Handbook of clinical neurology 2019 Jan 1 (Vol. 165, pp. 391-414). Elsevier.

The correct answer is: Rett syndrome

Question 69

Not answered

Marked out of 1

Which of the following disorders associated with intellectual disability is not a genetic deletion or duplication syndrome?

Select one:

- ☐ Klinefelter syndrome
- ☐ Smith-Magenis syndrome
- ☐ Di George (Velo-cardio-facial) syndrome
- ☐ Rubinstein-Taybi syndrome
- ☐ Cri-du-chat syndrome

Your answer is incorrect.

Explanation

Klinefelter syndrome (karyotype 47,XXY) is one of the most common chromosomal disorders. It is a sex chromosome trisomy.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 817.

The correct answer is: Klinefelter syndrome



Question 70

Not answered

Marked out of 1

Hypercalcemia with supra-ventricular aortic stenosis and elfin facial features are characteristic of

Select one:

- ☐ Cri du chat syndrome
- ☐ Smith-Magenis syndrome
- ☐ Di George syndrome
- ☐ Angelman syndrome
- ☐ Williams syndrome

Your answer is incorrect.

Explanation:

Clinical features of Williams syndrome include hypercalcemia (in 50%) with supra-ventricular aortic stenosis and elfin facial features. The syndrome is associated with anxiety and fears. Often there is abnormal attachment behaviour manifested as anxiety, poor peer relationships, and hypersensitivity (or conversely as social disinhibition and excessive friendliness). Verbal skills are often better than motor and visuospatial skills.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 86.

Sample D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 809.

The correct answer is: Williams syndrome

Question 71

Not answered

Marked out of 1

The presence of extraordinary proficiency in one isolated skill, despite intellectual disability in other domains, is called

Select one:

- ☐ Atypical asperger syndrome
- ☐ Heller syndrome
- ☐ Savant syndrome
- ☐ Spotted syndrome
- ☐ Isolated syndrome

Your answer is incorrect.

Explanation:

Of individuals with autism, a minority ('savants') may have 'islands of precocity' against a background of intellectual disability (i.e. isolated abilities, e.g. incredible memory or arithmetic skills).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 823.

The correct answer is: Savant syndrome

Question 72

Not answered

Marked out of 1

IQ assessment was carried out on a 7-year-old boy with a history of delayed developmental milestones. His IQ was calculated to be 56. What level of learning disability is he suffering from?

Select one:

- ☐ Profound LD
- ☐ Moderate LD
- ☐ Severe LD
- ☐ Mild LD
- ☐ Subnormal intelligence

Your answer is incorrect.

General intellectual functioning is determined by the use of standardized tests of intelligence, and the term significantly sub-average is defined as an IQ of approximately 70 or below or two standard deviations below the mean for the particular test. Mild LD has IQ range 50-69, Moderate with a range 35-49, Severe in the range 20-34, and Profound being below 20. Borderline intellectual functioning, according to DSM-5, is not within the diagnostic boundary of mental retardation and refers to a full-scale IQ in the 71 to 84.

The correct answer is: Mild LD

Question 73

Not answered

Marked out of 1

In males with the fragile X syndrome, which of the following statements is correct?

Select one:

- ☐ The delay in language development is usually unrelated to the degree of intellectual development
- ☐ The mean IQ score in adulthood is stable with increased age
- ☐ The mean IQ in affected boys is 20 to 30
- ☐ There is a relationship between the length of the DNA trinucleotide repeat and the level of intelligence
- ☐ Echolalia is rare

Your answer is incorrect.

Fragile X syndrome is an X-Linked disorder with an unusual pattern of inheritance. The fragility of the X chromosome is now known to be associated with an unstable region in the fragile-X mental retardation (FMR-1) gene, which was first described in 1991. This region of unstable DNA increases in length and degree of instability in successive generations (a pre-mutation) until a critical point is reached, and the gene no longer functions (a full mutation). The instability is caused by an increase in CGG repeats from the 50 or so repeats that are usual to 50-100 repeats (pre-mutation) to over 230 repeats (full mutation). The chance of a child inheriting a lengthened gene is proportional to the length of the unstable region in the carrier mother. The severity of the intellectual disability probably depends mostly on the proportion of cells in which the abnormal chromosome is inactivated, X inactivation being random. The associated intellectual disability is mild to moderate. (Ref: New Oxford textbook of psychiatry-2nd edition, 1840)

The correct answer is: There is a relationship between the length of the DNA trinucleotide repeat and the level of intelligence

Question **74**

Not answered

Marked out of 1

The Special Education Needs & Disability Act of 2001 (UK) requires schools to draw up accessibility strategies for children with intellectual disabilities. The aim of this programme is to facilitate

Select one:

- ☐ the inclusion of some pupils without intellectual disabilities in special needs schools
- ☐ the early provision of special needs education to pupils with intellectual disabilities
- ☐ the inclusion of pupils with intellectual disabilities in mainstream schools
- ☐ the enrolment of pupils with intellectual disabilities in special schools
- ☐ the identification of gifted pupils in mainstream schools and provide them with special education

Your answer is incorrect.

Explanation:

SENDA provisions increase access to a place in a mainstream school for children with special educational needs.

Ref: <https://adc.bmj.com/content/86/6/427>

The correct answer is: the inclusion of pupils with intellectual disabilities in mainstream schools



Question 75

Not answered

Marked out of 1

Which of the following statements about schizophrenia in individuals with intellectual disability is incorrect?

Select one:

- ☐ Patients are less likely to have complex delusional ideas
- ☐ The rate is raised threefold over the general population
- ☐ Thought disorder is probably more evident
- ☐ Age of onset tends to be earlier
- ☐ It is nearly always associated with a reduction in functional abilities

Your answer is incorrect.

Explanation:

Schizophrenia is approximately 3 times more common in those with intellectual disability than in the general population. Age of onset tends to be earlier (mean 23 years). It is more commonly associated with epilepsy, negative symptoms, and impairment of episodic memory. In severe ID there may be unexplained aggression, bizarre behaviour, labile mood, or increased mannerisms and stereotypies. Thought disorder and complex persecutory delusions are probably less evident.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 826.

The correct answer is: Thought disorder is probably more evident

Question 76

Not answered

Marked out of 1

Which of the following childhood disorders is more common in girls than in boys?

Select one:

- ☐ Tourette syndrome
- ☐ Asperger syndrome
- ☐ Conduct disorder
- ☐ ADHD
- ☐ Rett syndrome

Your answer is incorrect.

Explanation:

Rett syndrome is believed to be caused by a dominant X-linked gene. It is a degenerative disorder that almost exclusively affects females.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 103.

Persico AM et al. The psychopharmacology of autism spectrum disorder and Rett syndrome. In Handbook of clinical neurology 2019 Jan 1 (Vol. 165, pp. 391-414). Elsevier.

The correct answer is: Rett syndrome

Question 77

Not answered

Marked out of 1

A couple has a child with Down syndrome. Their risk of having another child with the same syndrome is highest if their first child has which of the following?

Select one:

- ☐ Trisomy 21 Down syndrome
- ☐ Trisomy 23 Down syndrome
- ☐ Mosaic Down syndrome
- ☐ Translocation Down syndrome
- ☐ Disomy 23 Down syndrome

Your answer is incorrect.

Explanation:

In translocation, a fusion occurs of two chromosomes, resulting in a total of 46 chromosomes despite the presence of an extra chromosome 21. The disorder, unlike trisomy 21, is usually inherited and the translocated chromosome may be present in unaffected parents and siblings. The asymptomatic carriers have only 45 chromosomes. When translocation is present, the risk of having a child with Down syndrome is about 1 in 3.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 101.

The correct answer is: Translocation Down syndrome

Question 78

Not answered

Marked out of 1

An autosomal recessive pattern of inheritance occurs in

Select one:

- ☐ Phenylketonuria
- ☐ Hunter disease
- ☐ Fragile X syndrome
- ☐ Tuberous sclerosis
- ☐ Lesch-Nyhan syndrome

Your answer is incorrect.

Explanation:

Phenylketonuria is a preventable cause of severe intellectual disability. It is an autosomal recessive syndrome characterised by deficits in phenylalanine hydroxylase enzyme activity which leads to a buildup of phenylalanine in the body.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 812.

The correct answer is: Phenylketonuria

Question 79

Not answered

Marked out of 1

A 32-year-old patient with intellectual disability is to be prescribed lithium for the treatment of his bipolar disorder. Which of the following baseline investigation must be done before initiating therapy?

Select one:

- ☐ Lipid profile
- ☐ Chest x-ray
- ☐ EEG
- ☐ ECG
- ☐ Urinalysis

Your answer is incorrect.

Explanation:

Given the increased prevalence of seizures in patients with intellectual disability (ID), an EEG is advised. Lithium use for people with epilepsy and ID is cautioned because of its potential to increase the frequency of convulsions at therapeutic doses, and because lithium has marked epileptogenic activity in overdose.

Ref: Royal College of Psychiatrists. Prescribing anti-epileptic drugs for people with epilepsy and intellectual disability. 2017.

The correct answer is: EEG

Question **80**

Not answered

Marked out of 1

Which of the following conditions is most commonly associated with atrioventricular septal defects in children?

Select one:

- ☐ Cri-du-chat syndrome
- ☐ Congenital hypothyroidism
- ☐ Benzodiazepine teratogenicity
- ☐ Valproate teratogenicity
- ☐ Down syndrome

Your answer is incorrect.

Explanation:

Congenital heart defects (CHDs) occur in approximately 50% of individuals with Down syndrome (DS), most commonly atrioventricular septal defect (42% of CHDs in individuals with DS), ventricular septal defect (22%) and atrial septal defect (16%).

Ref: Antonarakis SE et al. Down syndrome. Nature Reviews Disease Primers. 2020 Jan;6(1):9.

The correct answer is: Down syndrome

Question 81

Not answered

Marked out of 1

Which of the following is a common feature of Prader-Willi syndrome?

Select one:

- ☐ Violent self-harm
- ☐ Compulsive skin picking
- ☐ Near normal IQ in 50% of children
- ☐ Extreme shyness
- ☐ Paroxysmal laughter

Your answer is incorrect.

Explanation:

The cardinal features of Prader-Willi syndrome (PWS) include severe infantile hypotonia, hyperphagia with the onset of obesity during early childhood if not controlled, developmental delay with learning and behavioural problems, short stature with small hands/feet and hypogonadism/hypogenitalism due to growth hormone and other endocrine deficiencies. Mild craniofacial dysmorphism with enamel hypoplasia and a dry mouth are common. Psychiatric phenotypes, behavioural and autism features correlate with specific PWS genetic subtypes (e.g., autism in those with maternal disomy 15). Cognitive and behavioural problems (tantrums, compulsions, compulsive skin picking) are common. Global developmental delay and mental deficiency (average IQ = 65) occur in nearly all children with PWS.

Ref: Butler MG et al. Prader-Willi syndrome-clinical genetics, diagnosis and treatment approaches: an update. Current pediatric reviews. 2019 Nov 1;15(4):207-44.

The correct answer is: Compulsive skin picking



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